PROJECT REPORT

PREDICT NEW YORK CITY AIRBNB PRICES

Using Linear Regression

SUBMITTED BY

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ACADEMIC PROJECT • MACHINE LEARNING

PROJECT SUMMARY

Predict New York City Airbnb Prices Using Linear Regression

1. Objective

The objective of this project is to build a Linear Regression model that predicts Airbnb listing prices in New York City using available listing characteristics.

2. Data Preparation

The AB_NYC_2019 dataset initially contained 48,895 records and 16 columns. Missing values in *reviews_per_month* were filled with 0 because the corresponding listings had no reviews. The *last_review* column was removed, along with identifiers and text fields that were not used as model inputs. Extreme values in *price* and *minimum_nights* were filtered using their 95th-percentile thresholds ($355 and 30 nights). The resulting dataset contained 45,759 records. The high-cardinality *neighbourhood* column (219 unique categories) was removed, while *neighbourhood_group* and *room_type* were one-hot encoded with the first category dropped.

3. Methodology

The target variable was *price*, and the remaining 13 columns were used as features. The data was split into 80% training data (36,607 rows) and 20% testing data (9,152 rows), using *random_state=42*. A Linear Regression model was trained on the training set and evaluated on the test set.

4. Model Results

Metric	Result
R^2 score	0.471 (about 47.1% of test-set price variation explained)
Mean Absolute Error (MAE)	$38.54
Mean Squared Error (MSE)	2834.96
Root Mean Squared Error (RMSE)	$53.24

5. Residual Analysis and Findings

The actual-versus-predicted plot showed that predictions for higher-priced listings tended to remain below the ideal 45-degree line, indicating underprediction at the upper end of the price range. The mean absolute error was $29.03 for cheaper listings and $48.58 for expensive listings, so the model performed better on the cheaper group by this measure. The expensive-group MAE was approximately 67.4% higher.

6. Conclusion and Improvement

The Linear Regression model provides a useful baseline, but its predictive performance is moderate. The R^2 score of 0.471 indicates that important price variation remains unexplained. Future improvements could include retaining more location detail through suitable encoding, adding relevant property features, testing nonlinear models, and using cross-validation. These changes should be evaluated on held-out data.

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